EX. NO: 8

IMPLEMENTATION OF DIFFIE–HELLMAN KEY EXCHANGE ALGORITHM

AIM

To write a C program to implement the Diffie–Hellman Key Exchange Algorithm.

ALGORITHM

Step 1: Start the program.

Step 2: Read the public prime number P and primitive root G.

Step 3: Read the private keys of Alice (a) and Bob (b).

Step 4: Compute Alice's public key:

$$A = G^{a}\ P$$

Step 5: Compute Bob's public key:

$$B = G^{b}\ P$$

Step 6: Exchange the public keys.

Step 7: Alice computes the shared secret key:

$$K_A = B^{a}\ P$$

Step 8: Bob computes the shared secret key:

$$K_B = A^{b}\ P$$

Step 9: Verify that both shared keys are equal.

Step 10: Display the public keys and the shared secret key.

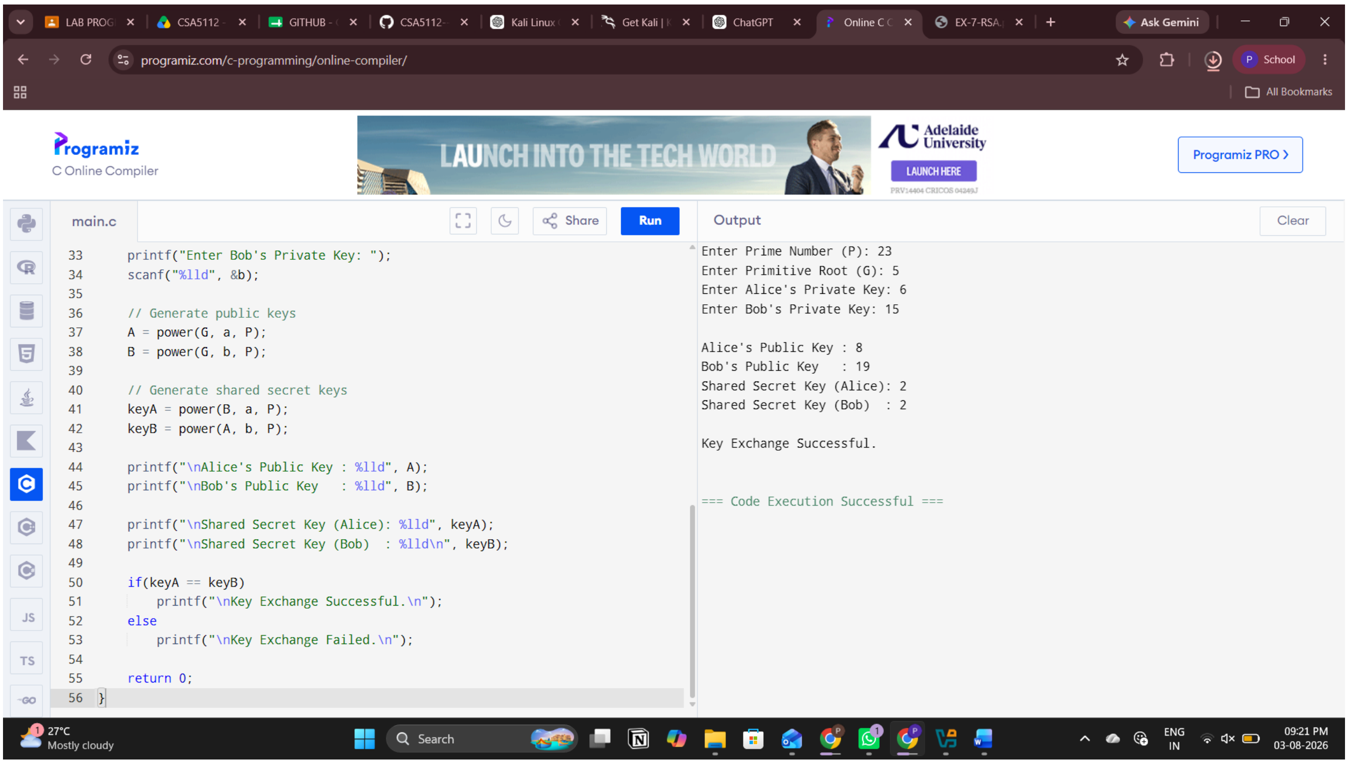
Step 11: Stop the program.

PROGRAM:

```
#include <stdio.h>

// Function to calculate (base^exp) % mod

long long power(long long base, long long exp, long long mod)

{

    long long result = 1;


    while(exp > 0)

    {

        result = (result * base) % mod;

        exp--;

    }
```

```
    return result;
}
int main()
{
    long long P, G;
    long long a, b;
    long long A, B;
    long long keyA, keyB;
    printf("Enter Prime Number (P): ");
    scanf("%lld", &P);
    printf("Enter Primitive Root (G): ");
    scanf("%lld", &G);
    printf("Enter Alice's Private Key: ");
    scanf("%lld", &a);
    printf("Enter Bob's Private Key: ");
    scanf("%lld", &b);
    A = power(G, a, P);
    B = power(G, b, P);
    keyA = power(B, a, P);
    keyB = power(A, b, P);
    printf("\nAlice's Public Key : %lld", A);
    printf("\nBob's Public Key   : %lld", B);
    printf("\nShared Secret Key (Alice): %lld", keyA);
    printf("\nShared Secret Key (Bob)  : %lld\n", keyB);
    if(keyA == keyB)
        printf("\nKey Exchange Successful.\n");
    else
        printf("\nKey Exchange Failed.\n");
    return 0;
}
```

OUTPUT:


programiz.com/c-programming/online-compiler/
Programiz
C Online Compiler
main.c
Share
Run
printf("Enter Bob's Private Key: ");
scanf("%lld", &b);
// Generate public keys
A = power(G, a, P);
B = power(G, b, P);
// Generate shared secret keys
keyA = power(B, a, P);
keyB = power(A, b, P);
printf("\nAlice's Public Key : %lld", A);
printf("\nBob's Public Key   : %lld", B);
printf("\nShared Secret Key (Alice): %lld", keyA);
printf("\nShared Secret Key (Bob) : %lld\n", keyB);
if(keyA == keyB)
printf("\nKey Exchange Successful.\n");
else
printf("\nKey Exchange Failed.\n");
return 0;
}
Output
Clear
Enter Prime Number (P): 23
Enter Primitive Root (G): 5
Enter Alice's Private Key: 6
Enter Bob's Private Key: 15
Alice's Public Key : 8
Bob's Public Key   : 19
Shared Secret Key (Alice): 2
Shared Secret Key (Bob) : 2
Key Exchange Successful.
=== Code Execution Successful ===